

Insights UI

The Integration Hub for Data Science and Artificial Intelligence at the University of Idaho

Infrastructure and Expertise provisioned by the Institute for Interdisciplinary Data Sciences (IIDS)

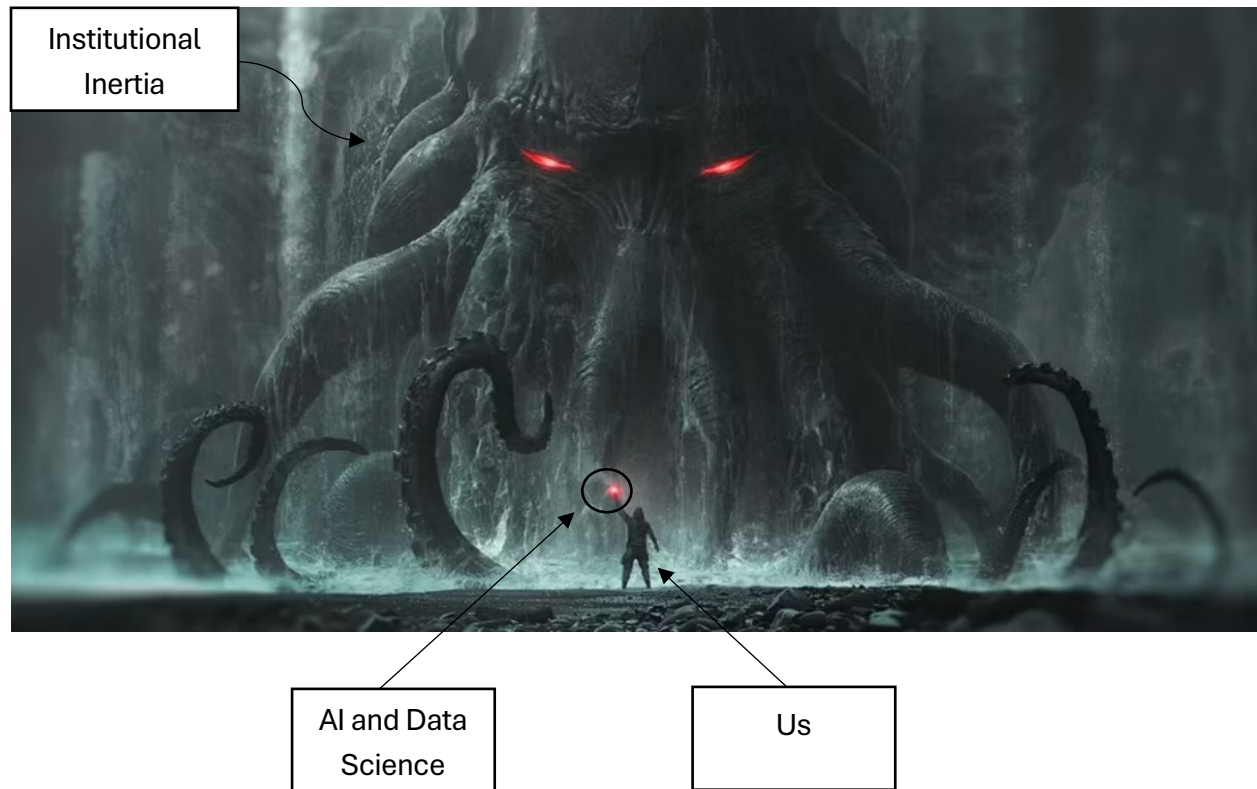


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Statement of Purpose and Current Activities

Purpose

To enhance research productivity and impacts at the University of Idaho by integrating data science and artificial intelligence (AI) solutions that support and advance the University's research ecosystem.

Needs Statement

The proliferation of data and the emergence of generative AI are fundamentally changing higher education. By democratizing access to data and developing trustworthy AI Tools, *Insights UI* will streamline administrative operations, elevate research excellence, unlock new opportunities for scientific discovery, facilitate collaboration with industry, and transform educational experiences. The University of Idaho has a choice: embrace the future and become a leader in this space or join the mass of universities scrambling to understand why they are getting left behind.

Activities and Services

Pillar 1: Build the DATA OBSERVATORY for the UI Research Enterprise: Democratize access to research administration data, enable data-driven decisions, and facilitate strategic analysis of strengths, opportunities, and competitive benchmarks.

Ongoing projects include:

- A. **Research Expenditures Dashboard.** An interactive real time dashboard of faculty activities, including research expenditures, scholarly outputs, and instruction. *(in collaboration with OSP, OIT, Library, and IR)*
- B. **OSP Analyst Workload Dashboard.** An interactive real time dashboard to track OSP analyst assignments, turnaround time, and workload. *(in collaboration with OSP)*
- C. **ORCID Integration.** A system to update UI faculty ORCID profiles using institutional data. *(in collaboration with OSP and the Library)*
- D. **VERSO Data Pipeline.** A system to integrate the VERSO RIMS with institutional data sources and applications. *(in collaboration with the Library)*

Pillar 2: Automate / Augment ORED Workflows with Generative AI: Leverage emerging approaches in generative AI to develop, deploy, and evaluate tools that automate manual processes, and augment the capabilities of research administration professionals.

Ongoing projects include:

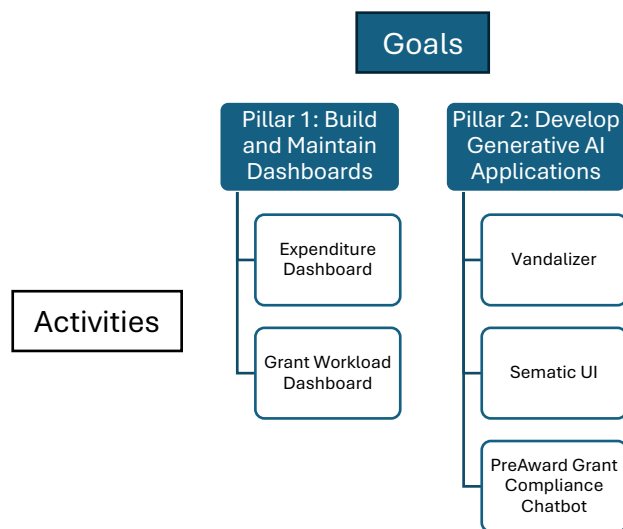
- A. **The Vandalizer:** AI powered document handling for extraction of structured data, completion of fillable pdfs, and administrative task augmentation. *(in collaboration with OSP and the Lawyers)*
- B. **Semantic UI:** An interactive agent built on the semantic network of UI faculty publications and research activities. *(in collaboration with the Library)*
- C. **Grant Compliance Chatbot:** A project specific chatbot that assimilates project documents and adapts its utility over the life cycle of a sponsored project. *(in collaboration with OSP)*
- D. **Postaward Email Handling:** AI enhanced automation of the postaward@uidaho.edu inbox to facilitate task assignment, data extraction, and process automation. *(in collaboration with OSP)*

Strategic Development and Priorities

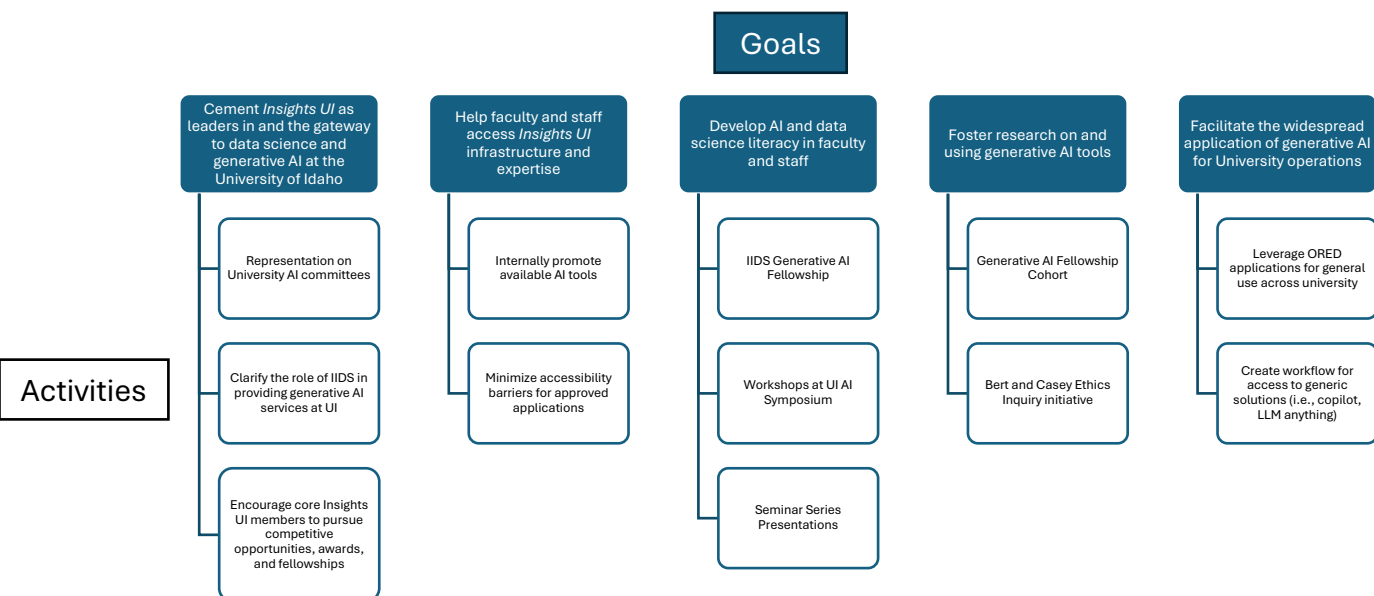
To meet the needs of the University of Idaho insert paragraph here about the need to develop infrastructure-physical and human, and deliver value-pillar 1 and 2, and research support. Initial value will be delivered through activities that align with Pillar's 1 and 2, and supporting research on and using generative AI. To maximize the capacity of Insights UI to deliver value, technical and human infrastructure needs to be developed. Below is a list of strategic priorities, goals, and activities undertaken to achieve each goal.

Deliver Value:

Strategy 1: Improve the Operational Capacity of the Office of Research and Economic Development

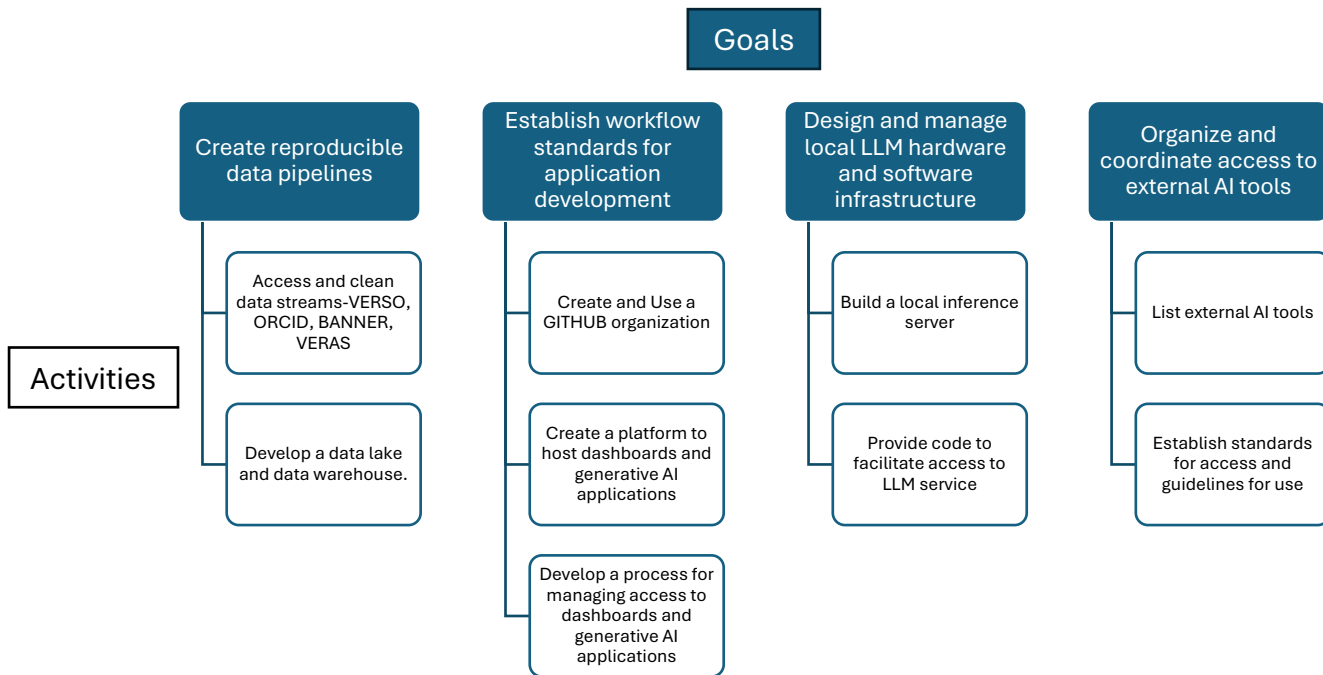


Strategy 2: Engage and Empower Faculty and Staff

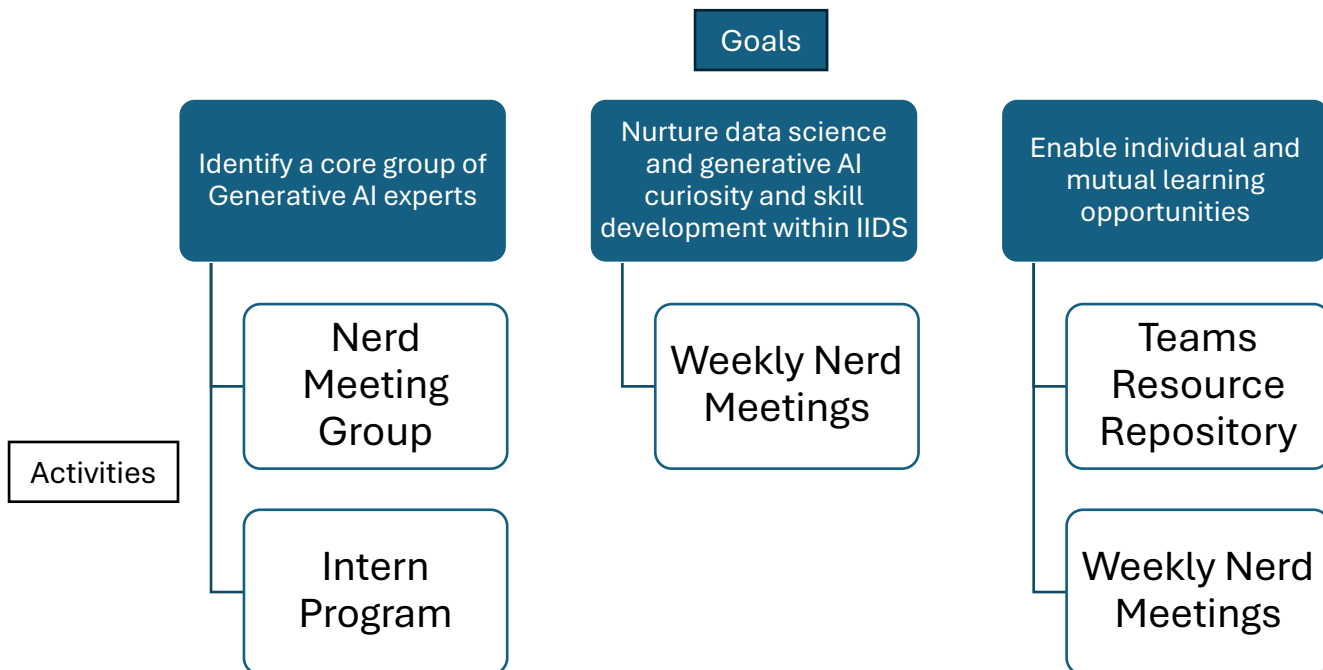


Infrastructure Development

Strategy 3: Establish Data Science and AI Technical Foundation



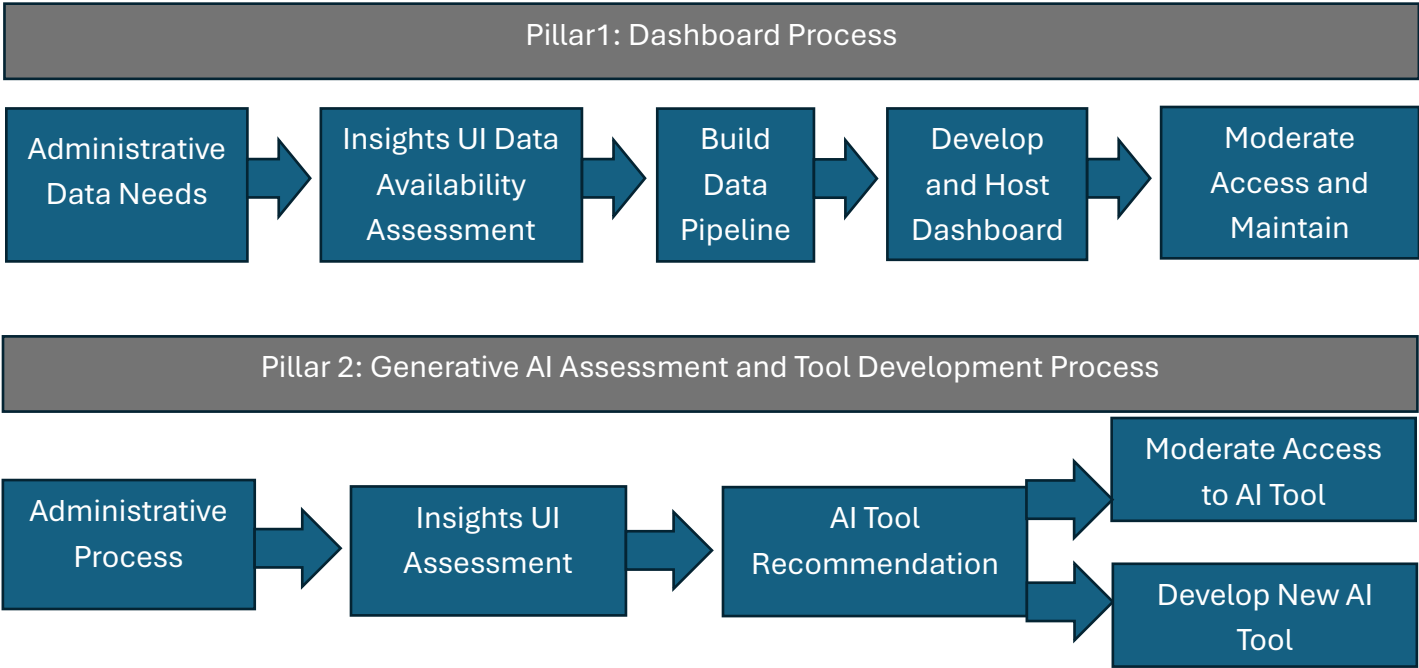
Strategy 4: Foster Data Science and AI Capabilities in IIDS



Insights UI: Generative AI and Dashboard

Assessment Process

Generative AI and the democratization of data present countless opportunities to revolutionize research and operations at the University of Idaho. The work of *Insights UI* has revealed numerous use cases beyond our current scope of work. However, IIDS staff are at capacity and deploying *Insights UI* beyond ORED will force delays in critical work and strain IIDS’ financial position. It is therefore crucial to develop clear processes and procedures that ensure timely, fair, robust, and standardized access to *Insights UI* services. Implementing the proposed process will help IIDS allocate its resources more efficiently, maximizing benefits for the University. Additionally, it will allow IIDS to develop adequate plans for meeting emerging infrastructure needs and leverage its expertise to lead the University's adoption of Generative AI and data-driven approaches.



Tool Recommendation Table

		AI Models	
		External (ex. Openai)	Local (ex. llama3)
Applications	Commercial	Activities: Access Moderation (OIT), Learning Resources, Recommendations Example: Chatgpt, Copilot	Activities: Inference Server Access, Learning Resources, Recommendations Example: Anything LLM
	Custom	Activities: Development, Access Moderation (Insights UI), Learning Resources Example: Vandalizer	Activities: Development, Inference Server Access, Learning Resources Example: Tickbase

Insight UI Activities for Assessment Processes

- Pillar 1: Dashboards
 - Client-IIDS Activities
 - Dashboard need assessment
 - Moderate access
 - Host dashboard on platform
 - Specific dashboard training
 - General data science literacy training
 - Evaluation of dashboard usage and feedback meetings for feature improvement
 - Internal Activities
 - Data availability assessment
 - Build data pipeline
 - Clean data
 - Design dashboard
 - Maintain and update dashboard
 - Develop generalized data science literacy training
- Pillar 2: Generative AI Tools
 - Client-IIDS Activities
 - Formalized assessment and recommendation process
 - Moderate access
 - General AI and data science literacy training
 - Specific AI application training
 - Evaluation of AI tool usage and feedback meetings for tool improvement
 - Internal Activities
 - Maintain list of commercial AI tools
 - Obtain external LLM access keys and develop guidelines for access
 - Host AI application platform for custom LLM driven applications
 - Build and maintain local inference server
 - Identify internal assessment group
 - Develop generalized AI/data science literacy training

Activity 2: Generative AI Research Fellowship

Generative AI is poised to fundamentally change the way scholars across all disciplines conduct research by improving scholarly workflows, enabling new methodologies, creating data, and providing access to AI Research Assistants. Despite the breadth of new and emerging opportunities, the generative AI learning curve is a barrier many scholars will need expert help navigating. To empower researchers at the University of Idaho to use Generative AI in their research, we propose a small-group, cohort-based fellowship program through IIDS to help fellows build their AI literacy and develop an AI research project. The fellowship will be one semester long and culminate in research presentations from each of the participants.

IIDS stands to benefit in many ways by facilitating this fellowship. First, this aligns with the vision of Insights UI to **develop** and **empower** scholars to use generative AI. Second, the fellowship provides an opportunity to establish ourselves as **leaders** in the Generative AI space on campus. Such a fellowship creates a series of highly visible accomplishments in the Generative AI space at the University of Idaho including, but not limited to, the fellowship itself, the curriculum for the fellowship, and the research projects that emerge from the fellowship. Third, we increase the long-term demand for IIDS infrastructure. By helping scholars develop Generative AI research projects, IIDS increases its potential user base. Fourth, this fellowship provides an opportunity for IIDS to assess the latent (i.e., unexpressed) needs of researchers and identify areas of potential internal investment.

Deliverables

Immediate

- Fellows
 - Research presentation
- IIDS
 - Seminar Series Presentations (i.e., from fellows)
 - Generative AI in research needs assessment
 - Generative AI curriculum for research
 - Ethics Workshop Plan
 - Ethics Workshop Broader Impact Boiler Plate Language

Long-Term

- IIDS
 - Builds scholars ability and likelihood to use IIDS infrastructure in their research and grants.
 - Generative AI literacy publications from fellows and leadership team.
 - Develops ethics infrastructure for future generative AI grants.

- Enables a forward-thinking approach to generative AI infrastructure development, which is necessary due to the lack of guidance and known/accepted practices.
- Facilitates cross-disciplinary collaborative opportunities

Insight UI: Vision Statement

Insights UI aspires to be an irreplaceable resource for faculty and research administration at the University of Idaho through the application of data science and generative AI.

We aim to become:

- **Leaders** in the area of data science and generative AI at the University of Idaho due to our expertise in these domains.
- **Innovators** who develop Data science and Generative AI applications for faculty and research administration.

We want our efforts to result in:

- **Felt Impact:** The unit aims to deliver tangible applications—not solely theoretical contributions—that make a difference to researchers and improve research administration efforts. This includes enabling research about or using generative AI.
- **Recognized Excellence:** The unit strives to be a **center of excellence** at the University of Idaho where University decision-makers understand the value of the unit's contributions.
- **Institutional Reputation:** A campus-wide, state, and national reputation for developing and applying data science and generative AI applications in support of research and research administration.

We want Insights UI to be characterized as:

- **Adaptable:** AI and data science are rapidly evolving technologies, and the unit should be flexible and ready to adapt through constant learning, experimental development efforts, and communication with research administration, faculty, and public/private sector practitioners.
- **Sustainable:** Stable **resources** are critical to developing a unit capable of supporting the University of Idaho's research and research administration efforts requires a flexible, innovative, and competent group of experts with the ability to acquire and maintain cutting-edge hardware and software.
- **Fulfilling:** The unit should be an intellectually and professionally fulfilling experience due to the focus on interesting and cutting-edge work, emphasis on constant internal development, and an organizational culture that recognizes and values each member's contributions.

Activities of Insights UI will focus on:

- **Discovery:** Our efforts are motivated by a desire to *learn* about data science and generative AI, new solutions to research administration problems, and how to apply data science and generative AI to research scholarly topics.
- **Development:** Our efforts should focus on the ultimate goal of developing tangible applications using data science and generative AI. The application of data science and AI expertise promotes discovery that has a felt impact.
- **Empowerment:** Our efforts to *discover* and *develop* put us in a unique position to help engage and empower faculty and research administration to fully utilize the unit's data science and AI infrastructure.

Appendix: Assessment Template Draft:

1. Task Analysis

1.1. Task Overview

- Provide a concise description of the task/process
- List key deliverables or goals

1.2. Data Requirements

- Describe the INTERNAL data required by the project
 - a. Location, format, volume (amount), velocity (static, real time, etc), veracity (data quality)
 - b. What are the required data security protocols? Consider FERPA, CUI, proprietary information, etc.
- Describe the EXTERNAL data required by the project
 - a. Location, format, volume (amount), velocity (static, real time, etc), veracity (data quality)
 - b. What are the required data security protocols? Consider FERPA, CUI, proprietary information, etc.
 - c. What are the costs, licenses, MOUs, NDAs, etc?

1.3. AI Potential

- Describe the envisioned role of AI in this task/process
- Identify relevant AI capabilities:
 - a. Content generation (text, audio, visual)
 - b. Information extraction
 - c. Summarization
 - d. Insights/recommendations based on reference documents
 - e. Other (specify)

2. AI Applicability

2.1. Current Automation Level

- Estimate the percentage of the task that can be automated without AI

2.2. AI Value Proposition

- Outline the potential benefits of AI implementation

2.3. Development Complexity

- Assess the difficulty of developing a specific use-case application

2.4. User Adoption

- Evaluate the need for changes in user behavior, and the scope of required training

3. AI Model/Tool Selection

3.1. Model Considerations

- Data security/privacy requirements: Yes / No

- Accuracy comparison: Local vs. External models
- Cost-benefit analysis of using external models

3.2. Tool Evaluation

- Assess existing tools:
 - c. Commercial (e.g., ChatGPT, Anything LLM)
 - d. Custom (e.g., Vandalizer)
- Review similar tools in Insights UI

4. Insights UI Capacity Assessment

4.1. Development Capacity

- Evaluate Insights UI's ability to develop a custom tool

4.2. Skill Development

- Identify new skills required/acquired through tool development

4.3. Maintenance Requirements

- Estimate ongoing maintenance costs and expectations

4.4. User Training

- Assess the need for Insights UI to train users: Yes / No

4.5. Cost Analysis

- Client interactions
- Software R&D
- Ongoing maintenance
- Hardware utilization
- Feedback and evaluation processes

4.6. Benefit Analysis

4.6.1. Client Benefits

- Quality and quantity improvements in deliverables
- Task efficiency enhancements
- Potential as an institutional force multiplier

4.6.2. Insights UI Benefits

- Revenue potential
- New skills and capabilities acquisition
- Brand enhancement

5. Risk Assessment

5.1. Identify potential social/technical/economic risks associated with AI implementation

5.2. Propose mitigation strategies for each identified risk

6. Implementation Timeline

6.1. Estimated development time

6.2. Key milestones and deliverables

7. Success Metrics

7.1. Define quantifiable metrics to measure the success of AI implementation

7.2. Establish a timeline for reviewing and assessing these metrics

8. Stakeholder Feedback

8.1. Identify key stakeholders

8.2. Plan for gathering and incorporating stakeholder input throughout the process